Landscaping prognostic gene expression and anticancer targets using sample- and lineage-consensus scores derived from pan-cancer omics data

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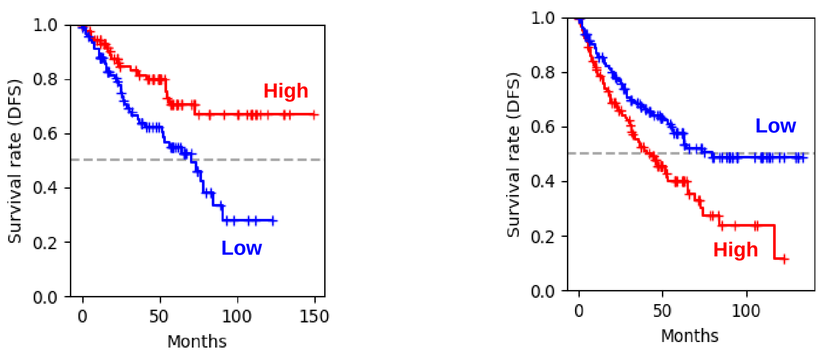
Abstract

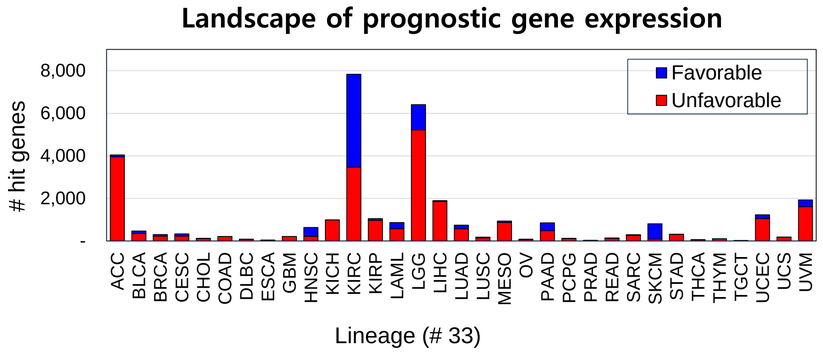
Integration of large scale multi-omics data enables extensive analysis of cross-associated data pairs between collateral datasets such as mutation, RNA/protein expression, RNAi/drug screen, clinical information, etc. Here, we investigated the reproducibility of the significance in the association by measuring sample consensus and/or lineage consensus for given data pairs (gene expression vs. patient survival, gene expression vs. RNAi efficacy). The sample consensus score represents the number of significant (p<0.01) cases in the associations for a given data pair among varied sample groups within a lineage, while the lineage consensus score is the significant cases among diverse lineages. ~74% of prognostic gene expression (i.e., gene expression associated with patient survival) were exclusively found in a single lineage, while only ~1.8% prognostic genes showed the significance in over three different lineages (consensus score >3). In case of sample consensus, common prognostic genes between genders were less than exclusive genes between them. However, prognostic genes with high sample consensus score generally exhibited high lineage consensus scores, implying that consensus scoring might improve the reproducibility of predicted prognostic gene expression in other cohort or cancer lineages. Furthermore, the lineage consensus score for the association between sgRNA (CRISPR) efficacy and RNA expression could predict the association between shRNA and RNA expression. This study suggests that consensus scoring help smart data mining prioritizing true-positives for the further validation. The concept of consensus scoring has been implemented in Q-omics software (http://qomics.sookmyung.ac.kr) for analyzing associations between diverse cancer omics data.

Ⅰ. Data mining for survival analysis

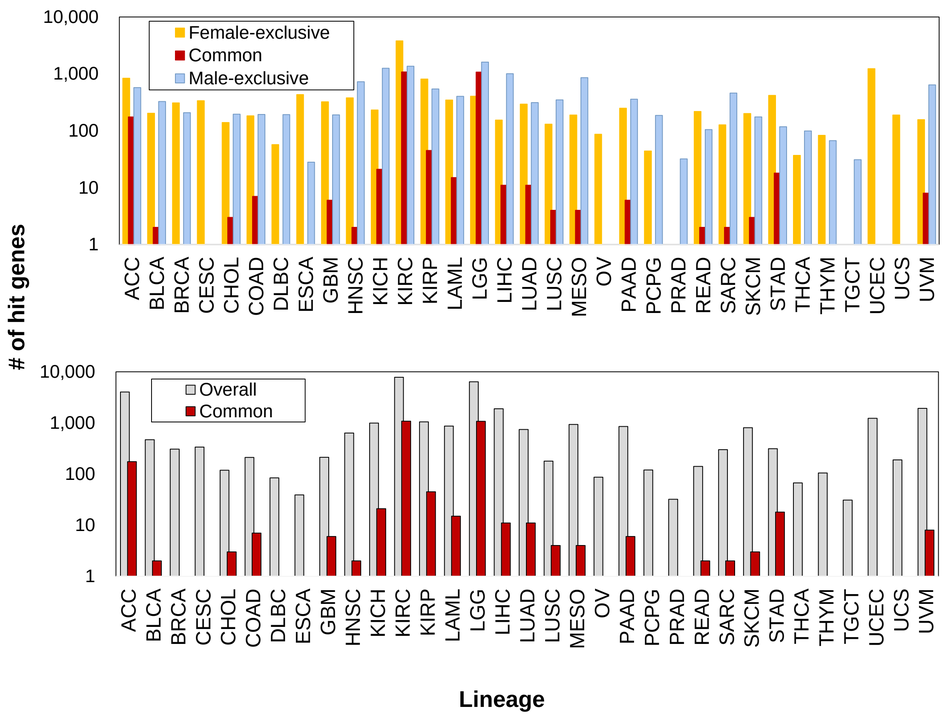
Favorable prognostic gene expression **Unfavorable prognostic gene expression**



Landscape of prognostic gene expression

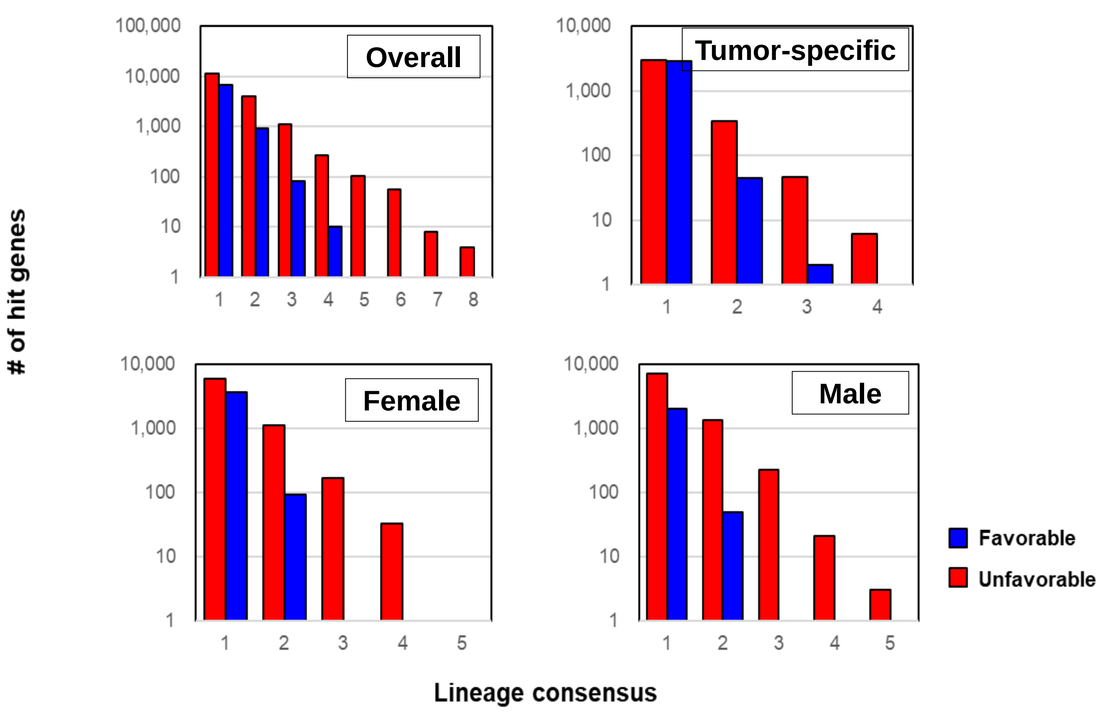


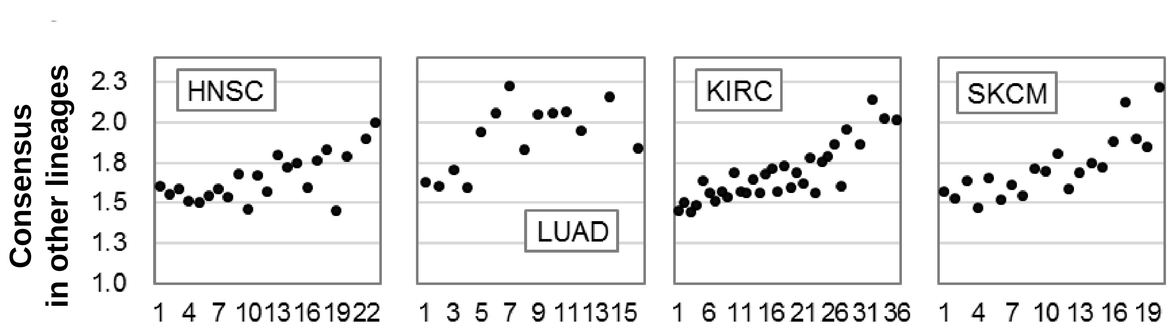
Landscape of gender-prognostic gene expression (Log scale)



We performed a large-scale analysis using public data to find new prognostic gene expression and anticancer targets for diverse cancers. Moreover, we defined the favorable and unfavorable prognostic gene expression. Favorable prognostic gene expression is a lower survival rate in the group with low gene expression, and unfavorable prognostic gene expression is a lower survival rate in the group with high gene expression. Distribution of prognostic gene expression shows that there is difference in the number depending on the cancer lineage. In addition, low gender consensus was confirmed by the large difference between gender-common and overall prognostic gene. Therefore, we analyzed sample- and lineage-consensus.

Ⅱ. Sample- and lineage-consensus

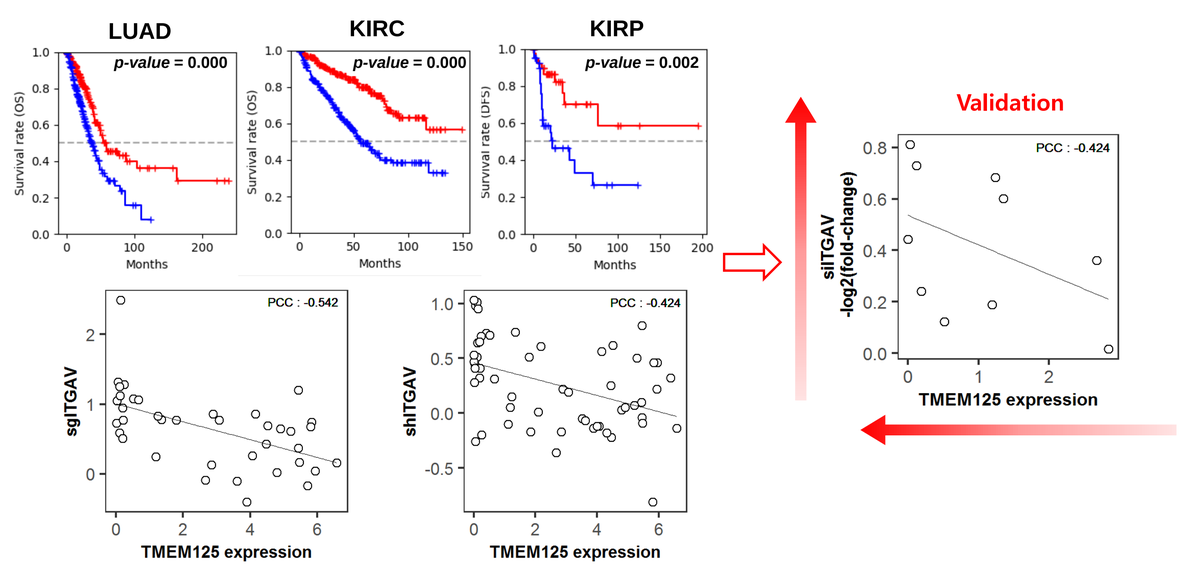




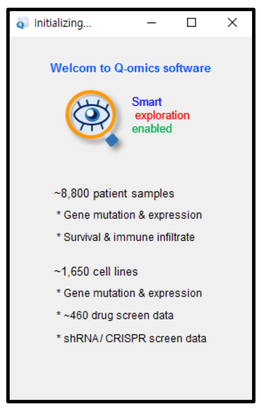
Consensus among samples within the lineage

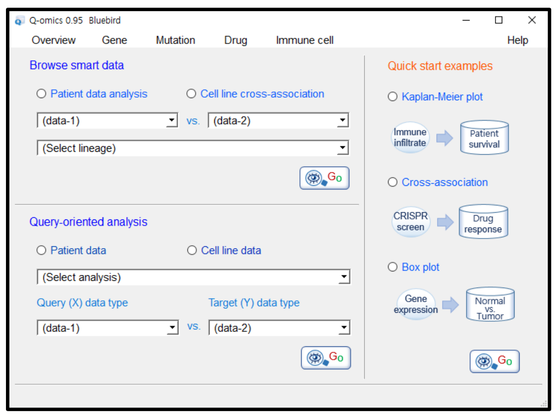
A consensus scoring method was used to preferentially select data with high experimental validation success rates. The "Sample Consensus Score" is derived by iteratively calculating the significance (*p-value*) of the correlation between gene expression and patient survival in a specific cancer type through various sampling methods, and the significance of several other cancer lineages is derived from the "Lineage consensus score". we compared the two consensus scores. The two consensus scores were highly correlated. This means that the predictability is high, and we thought this was an indicator that could increase the experimental success rate.

Ⅲ. Validation of consensus scoring



Q-omics software





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Reference

Q-omics: Smart Software for Assisting Oncology and Cancer Research, Jieun Lee, Youngju Kim, Seonghee Jin, Heeseing Yoo, Sumin Jeong, Euna Jeong and Sukjoon Yoon, Molecules and Cells, 2021 Nov 30;44(11):843-850.